

Moving-Frame Memory-Tail Recurrences

Exact Birth Slices, Sharp Defect Forcing, and Strong Raw Contraction

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Abstract

RH-133 fixed a natural dyadic packet gauge but found that packet angles do not control normalized tail alignment. We now return to the source of the tail and derive its exact one-step law. For trace-normalized snapshots $X_t \succeq 0$, memory decay $0 \leq \eta < 1$, and depth d , write

$$T_t = \sum_{a=d}^t \eta^a X_{t-a}.$$

Then

$$T_{t+1} = \eta T_t + \eta^d X_{t-d+1}.$$

The second term is the unique snapshot crossing the finite-memory boundary.

If U_t, U_{t+1} are moving four-frames, O_t is their polar alignment, $E_t = U_{t+1} - U_t O_t$, and $K_t = U_t^* T_t U_t$, the exact identity and the sharp matrix Young inequality imply, for every $\tau > 0$,

$$K_{t+1} \preceq \eta(1 + \tau) O_t^* K_t O_t + \eta(1 + \tau^{-1}) \|T_t\| \|E_t\|^2 I + \eta^d U_{t+1}^* X_{t-d+1} U_{t+1}.$$

Thus the raw compressed tail has a source-level affine recurrence: old-tail transport, frame-change forcing, and a positive birth slice.

For the finite tail envelope $\delta_p = \eta^d(1 - \eta^p)/(1 - \eta)$, its ratio is at most $1 + \eta$. Consequently the weighted tail $D_t = \delta_p K_t$ has old-tail coefficient at most $\eta(1 + \tau)(1 + \eta)$. With the model values $\eta = 1/512$ and $\tau = 1$, this is 0.0039138794.

All 330 temporal transitions in the five-scale packet satisfy the exact identity and both raw and weighted Loewner bounds. The maximum identity error is 2.62×10^{-30} ; 24 transitions are first births. Although consecutive frames can be nearly orthogonal, every one of the 240 nonzero forcing updates is dominated by the boundary-slice term. Strong raw contraction is therefore proved; the remaining difficulty is conversion to the recent-Gram relative metric.

1 Finite memory separates old tail from new birth

Let the normalized memory Gramian satisfy

$$M_t = X_t + \eta M_{t-1}, \quad \text{Tr } X_t = 1,$$

with $M_{-1} = 0$. Split it at depth d into

$$R_t = \sum_{a=0}^{d-1} \eta^a X_{t-a}, \quad T_t = \sum_{a=d}^t \eta^a X_{t-a},$$

where absent negative-time snapshots are zero. Both pieces are positive, and $M_t = R_t + T_t$. Direct assembly of these disjoint sums was already essential in RH-130 because it avoids cancellation between nearly equal full and recent Gramians.

Theorem 1 (Exact tail-birth identity). *For every t ,*

$$T_{t+1} = \eta T_t + B_{t+1}, \quad B_{t+1} = \eta^d X_{t-d+1} \succeq 0.$$

At the first crossing of the memory boundary, $T_t = 0$ and the entire next tail is the birth slice B_{t+1} .

Proof. Shift the age index in ηT_t . Every old term in T_{t+1} is obtained, except the age- d term $\eta^d X_{t-d+1}$. Positivity is inherited from the snapshot. $\square$

The identity explains the rank-creation obstruction of RH-130 without any spectral calculation. A zero tail becomes nonzero because a positive snapshot crosses the boundary. No finite multiplicative factor can create that range; the correct recurrence is affine from the start.

2 Moving-frame compression

Let U_t and U_{t+1} be isometries from $\mathbb{C}^r$ into the common source-coordinate space. Choose an orthogonal O_t and write

$$U_{t+1} = U_t O_t + E_t.$$

The polar/Procrustes choice minimizes the Frobenius defect among orthogonal maps [Golub and Loan, 2013]. Define the compressed old tail $K_t = U_t^* T_t U_t$.

Theorem 2 (Moving-frame affine tail bound). *For every $\tau > 0$,*

$$\begin{aligned} K_{t+1} &\preceq \eta(1 + \tau) O_t^* K_t O_t \\ &\quad + \eta(1 + \tau^{-1}) \|T_t\| \|E_t\|^2 I \\ &\quad + U_{t+1}^* B_{t+1} U_{t+1}. \end{aligned}$$

The coefficients $1 + \tau$ and $1 + \tau^{-1}$ are jointly sharp under only norm information on the frame defect.

Proof. Insert Theorem 1 into $K_{t+1} = U_{t+1}^* T_{t+1} U_{t+1}$. For the old-tail term set $A = T_t^{1/2} U_t O_t$ and $C = T_t^{1/2} E_t$. The positive matrix form of Young's inequality gives

$$(A + C)^*(A + C) \preceq (1 + \tau) A^* A + (1 + \tau^{-1}) C^* C.$$

Moreover $C^* C \preceq \|T_t\| \|E_t\|^2 I$. Multiplication by η and addition of the birth compression prove the formula. In the scalar case, $C = \tau A$ attains equality in Young's inequality, proving joint sharpness [Bhatia, 1997]. $\square$

Large frame motion does not enter the multiplicative coefficient; it enters as positive forcing weighted by the already decayed old-tail norm. This is the structural advantage of deriving the recurrence before normalizing by a possibly weak recent Gramian.

3 Finite-envelope weighting

Since every snapshot has trace one,

$$\|T_t\| \leq \text{Tr } T_t \leq \delta_p = \eta^d \frac{1 - \eta^p}{1 - \eta},$$

where $p = t - d + 1$ is the number of discarded snapshots. For $p \geq 1$,

$$\frac{\delta_{p+1}}{\delta_p} = \frac{1 - \eta^{p+1}}{1 - \eta^p} \leq 1 + \eta,$$

with equality at $p = 1$.

Corollary 3 (Weighted affine recurrence). *Set $D_t = \delta_p K_t$. Away from the first birth,*

$$D_{t+1} \preceq \rho_p O_t^* D_t O_t + F_t,$$

where

$$\rho_p = \eta(1 + \tau) \frac{\delta_{p+1}}{\delta_p} \leq \eta(1 + \tau)(1 + \eta)$$

and

$$F_t = \delta_{p+1} \left[\eta(1 + \tau^{-1}) \delta_p \|E_t\|^2 I + U_{t+1}^* B_{t+1} U_{t+1} \right].$$

At the first birth, take $\rho_p = 0$ and the whole next weighted tail as forcing.

The first-birth convention is not a weakness: it is the exact semidefinite logic required by RH-130. For $\eta = 1/512$ and $\tau = 1$, the uniform old-tail coefficient equals

$$\frac{2}{512} \left(1 + \frac{1}{512} \right) = 0.00391387939453125.$$

The parameter τ can later be optimized against the relative size of frame forcing, but no optimization is needed to expose strict contraction.

4 Five-scale temporal audit

The audit uses five scales, two channels, three adaptive thresholds, and all consecutive times from the first four-frame action to the inherited endpoint. This gives 30 temporal chains and 330 transitions. Recent and old memory parts are assembled directly. At each step the two packet input frames are polar aligned, the exact boundary snapshot is identified, and the raw and weighted Loewner slacks are evaluated.

All 330 exact tail identities pass; the maximum spectral-norm discrepancy is 2.62×10^{-30} . Every raw and weighted Loewner comparison passes. There are 24 first-birth transitions, one

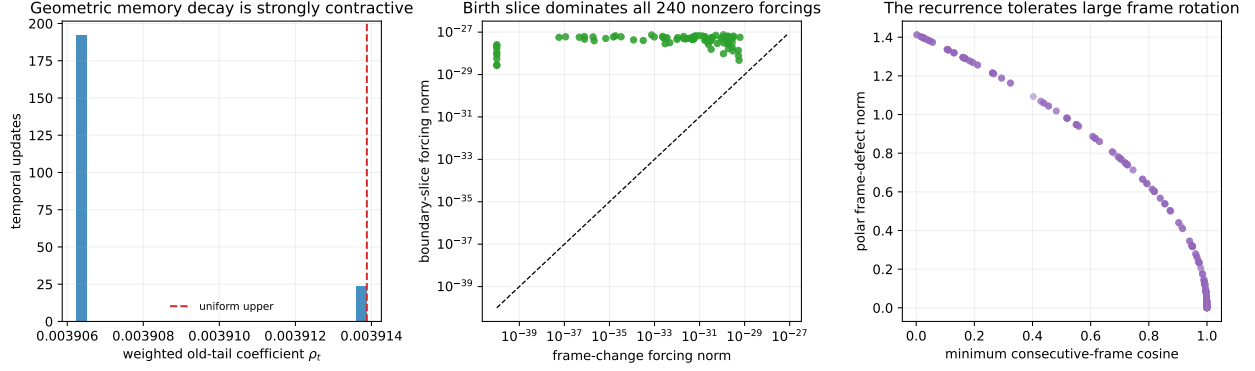


Figure 1: Strong weighted contraction, birth versus frame forcing, and consecutive-frame motion across all 330 temporal updates.

in each eligible scale/channel/threshold chain. The envelope ratio is reproduced exactly in floating arithmetic and the maximum ρ_p is the theoretical 0.0039138794.

Frame coherence is not assumed. The minimum principal cosine is 0.002204, the median spectral frame defect is 0.567, and the maximum is 1.413. Nevertheless all 240 updates with nonzero forcing are birth-dominated: the boundary-slice norm exceeds the frame-forcing norm in every case. Weighted forcing norms lie between 2.69×10^{-29} and 7.53×10^{-28} , with median 4.92×10^{-28} .

Those tiny absolute numbers are not yet a relative Rayleigh conclusion. The recent action Gramian can have condition number near 10^{10} , so normalization by $G_{t+1}^{-1/2}$ may amplify forcing severely. This is the precise remaining bridge, not a defect in the raw memory recurrence.

5 Route consequence and boundary

RH-134 proves the source mechanism sought after RH-130: old tail contracts by the geometric memory factor, rank is created only by an explicit positive boundary slice, and moving-frame error is a separately quantified forcing. The result is stronger than a numerical fit and independent of the Gaussian details beyond snapshot positivity and trace normalization.

The next layer must transport the recent Gram metric simultaneously. It should convert Corollary 3 into $\gamma_{t+1}^2 \leq \rho_t \gamma_t^2 + q_t$, determine how weak Gram directions amplify the two forcing pieces, and isolate a sharp obstruction if no uniform lower metric bound is available.

We have not proved a relative-Gram recurrence, a cross-scale source identity, uniform bounds on q_t , a normalized-base liminf, uniform Stage A, a Hilbert–Polya operator, zeta-zero identification, or the Riemann Hypothesis.

References

- Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.
- Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, 4 edition, 2013.